

Functional Requirements:

1. The system shall be able to receive and read a file supplied by the user from the command line.
2. The system shall be able to check the file for correct formatting of information and return an error if necessary.
3. The system shall be able to store user submitted values into a location in memory starting from memory location 00 and increment by one.
4. The system shall accept inputs from users and provide feedback as to if the number provided is in range.
5. The system shall have an accumulator that is loaded with no value until other functions access and store values in the accumulator.
6. The system shall have a Program Counter, so the operations of the file are put in correct order.
7. The system shall add the value at a memory address to the accumulator when it hits an ADD instruction (opcode 30).
8. The system shall subtract the value at a memory address from the accumulator when it hits a SUBTRACT instruction (opcode 31).
9. The system shall multiply the accumulator by the value at a memory address when it hits a MULTIPLY instruction (opcode 33).
10. The system shall divide the accumulator by the value at a memory address when it hits a DIVIDE instruction (opcode 32), but only after checking the divisor isn't zero.
11. The system shall set the program counter to a new address when it hits a BRANCH instruction (opcode 40).
12. The system shall be able to branch to a specific memory location if the accumulator is a negative value (opcode 41).
13. The system shall be able to branch to a specific memory location if the accumulator value is zero (opcode 42).
14. The system shall be able to handle overflows via truncation to keep the last 4 digits cutting numbers off that are higher or lower than 9999 and -9999.
15. The system shall stop running the program when it hits a HALT instruction (opcode 43).

Non-Functional Requirements:

1. The system shall have a GUI that is colorblind accessible.
2. The system will always display values as they are stored to help users understand the way in which the values are stored.
3. The system shall run using just Python 3, without needing the user to install any extra packages.